

# Content

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## Externally assessed

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### Description

The Pearson Edexcel International GCSE in Further Pure Mathematics requires students to demonstrate application and understanding of the following:

#### **Number**

- Use numerical skills in a purely mathematical way and in real-life situations.

#### **Algebra and calculus**

- Use algebra and calculus to set up and solve problems.
- Develop competence and confidence when manipulating mathematical expressions.
- Construct and use graphs in a range of situations.

#### **Geometry and trigonometry**

- Understand the properties of shapes, angles and transformations.
- Use vectors and rates of change to model situations.
- Use coordinate geometry.
- Use trigonometry.

Students will be expected to have a thorough knowledge of the content common to the Pearson Edexcel International GCSE in Mathematics (Specification A) (Higher Tier) or Pearson Edexcel International GCSE in Mathematics (Specification B).

Questions may be set which assumes knowledge of some topics covered in these specifications, however knowledge of statistics and matrices will not be required.

Students will be expected to carry out arithmetic and algebraic manipulation, such as being able to change the subject of a formula and evaluate numerically the value of any variable in a formula, given the values of the other variables.

The use and notation of set theory will be adopted where appropriate.

## Assessment information

Each paper is assessed through a 2-hour examination set and marked by Pearson.

The total number of marks for each paper is 100.

Each paper will consist of around 11 questions with varying mark allocations per question, which will be stated on the paper.

Each paper will contain questions from any part of the specification content, and the solution of any questions may require knowledge of more than one section of the specification content.

Each paper will have approximately 40% of the marks distributed evenly over grades 4 and 5 and approximately 60% of the marks distributed evenly over grades 6, 7, 8 and 9.

Diagrams will not necessarily be drawn to scale and measurements should not be taken from diagrams unless instructions to this effect are given.

A formulae sheet (*Appendix 4*) will be included in the written examinations.

A calculator may be used in the examinations (please see *page 22* for further information).

Questions will be set in SI units (international system of units).

# 1 Logarithmic functions and indices

| What students need to learn                                                                  | Notes                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> The functions $a^x$ and $\log_b x$ (where $b$ is a natural number greater than one) | A knowledge of the shape of the graphs of $a^x$ and $\log_b x$ is expected, but not a formal expression for the gradient.                                                                                                                                                                                                                                 |
| <b>B</b> Use and properties of indices and logarithms, including change of base              | <p>To include:</p> $\log_a xy = \log_a x + \log_a y,$ $\log_a \frac{x}{y} = \log_a x - \log_a y,$ $\log_a x^k = k \log_a x,$ $\log_a a = 1$ $\log_a 1 = 0$ <p>The solution of equations of the form <math>a^x = b</math>.</p> <p>Students may use the change of base formulae:</p> $\log_a x = \frac{\log_b x}{\log_b a}$ $\log_a b = \frac{1}{\log_b a}$ |
| <b>C</b> Simple manipulation of surds                                                        | <p>Students should understand what surds represent and their use for exact answers.</p> <p>Manipulation will be very simple. For example:</p> $5\sqrt{3} + 2\sqrt{3} = 7\sqrt{3}$ $\sqrt{48} = 4\sqrt{3}$                                                                                                                                                 |
| <b>D</b> Rationalising the denominator                                                       | $10 \times \frac{1}{\sqrt{5}} = 2\sqrt{5} \text{ or } \frac{1}{2 - \sqrt{3}}$                                                                                                                                                                                                                                                                             |